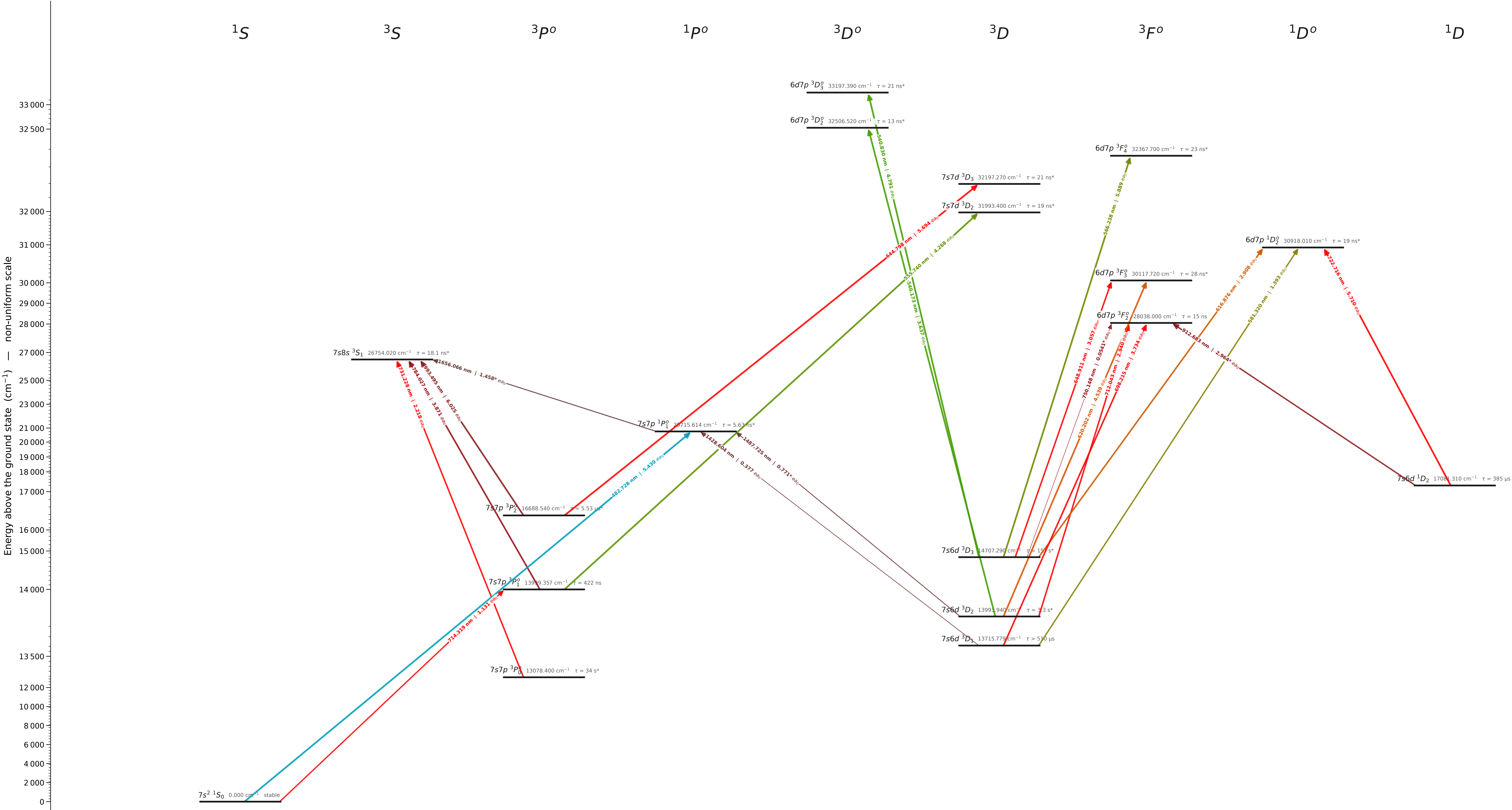
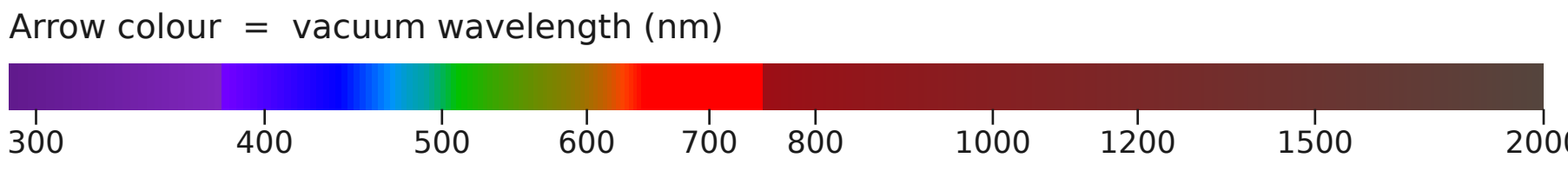


# Radium

Ra (Ra I). Strongest classified transitions below 2 μm (NIST): 18 levels, 22 lines.

Ra<sup>+</sup> ionisation limit 42573.36 cm<sup>-1</sup> = 5.27842 eV (234.889 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $|⟨J||er||J'⟩|$  in ea<sub>0</sub>  
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                                                          | λ vac (nm) | λ air (nm) | ν (THz)        | ⟨J  er  J'⟩  (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                          |
|---------------------------------------------------------------------|------------|------------|----------------|---------------------------------|----------------------|----------------|------------|--------------------------------------------------------------|
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.7277   | 482.5928   | 621.038489(15) | 5.430                           | 1.770e+08            | 28.27          | 99.7       | strong singlet line                                          |
| 7s6d 3D <sub>3</sub> – 6d7p 3F <sub>4</sub> <sup>o</sup>            | 566.2383   | 566.0812   | 529.4458       | 5.889                           | 4.300e+07            | 6.92           | 98.9       | 566.2 nm                                                     |
| 7s6d 3D <sub>1</sub> – 6d7p 1D <sub>2</sub> <sup>o</sup>            | 581.3199   | 581.1588   | 515.7099       | 1.393                           | 4.000e+06            | 8.377          | 7.6        | 581.3 nm                                                     |
| 7s6d 3D <sub>3</sub> – 6d7p 1D <sub>2</sub> <sup>o</sup>            | 616.8757   | 616.7051   | 485.9852       | 2.908                           | 1.460e+07            | 8.377          | 27.7       | 616.9 nm                                                     |
| 7s6d 3D <sub>2</sub> – 6d7p 3F <sub>3</sub> <sup>o</sup>            | 620.2020   | 620.0304   | 483.3788       | 4.539                           | 2.500e+07            | 5.684          | 70         | 620.2 nm                                                     |
| 7s7p 3P <sub>2</sub> <sup>o</sup> – 7s7d 3D <sub>3</sub>            | 644.7981   | 644.6200   | 464.9400       | 5.694                           | 3.500e+07            | 7.579          | 73.5       | 644.8 nm line used for collinear laser spectroscopy at IS... |
| 7s6d 3D <sub>3</sub> – 6d7p 3F <sub>3</sub> <sup>o</sup>            | 648.9112   | 648.7319   | 461.9931       | 3.057                           | 9.900e+06            | 5.684          | 27.7       | 648.9 nm                                                     |
| 7s6d 3D <sub>1</sub> – 6d7p 3F <sub>2</sub> <sup>o</sup>            | 698.2151   | 698.0226   | 429.369773(15) | 3.734                           | 1.660e+07            | 10.61          | 31         | proposed spin-flip repump channel for the blue-slower sch... |
| 7s6d 3D <sub>2</sub> – 6d7p 3F <sub>2</sub> <sup>o</sup>            | 712.0434   | 711.8472   | 421.031151(15) | 2.940                           | 9.700e+06            | 10.61          | 64         | proposed spin-flip repump channel for the blue-slower sch... |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 3P <sub>1</sub> <sup>o</sup> | 714.3185   | 714.1217   | 419.69016(3)   | 1.131                           | 2.370e+06            | 0.3771         | 100        | intercombination line                                        |
| 7s6d 1D <sub>2</sub> – 6d7p 1D <sub>2</sub> <sup>o</sup>            | 722.7157   | 722.5165   | 414.8138       | 5.710                           | 3.500e+07            | 8.377          | 66.5       | 722.7 nm                                                     |
| 7s7p 3P <sub>0</sub> <sup>o</sup> – 7s8s 3S <sub>1</sub>            | 731.2283   | 731.0269   | 409.9848       | 2.218                           | 8.500e+06            | 8.793          | 15.2       | 731.2 nm                                                     |
| 7s6d 3D <sub>3</sub> – 6d7p 3F <sub>2</sub> <sup>o</sup>            | 750.1476   | 749.9411   | 399.6446       | 0.054*                          | 2.815e+03            | 10.61          | 0.4        | 750.1 nm                                                     |
| 7s7p 3P <sub>1</sub> <sup>o</sup> – 7s8s 3S <sub>1</sub>            | 784.0270   | 783.8113   | 382.3752       | 3.871                           | 2.100e+07            | 8.793          | 38.3       | 784.0 nm                                                     |
| 7s6d 1D <sub>2</sub> – 6d7p 3F <sub>2</sub> <sup>o</sup>            | 912.6828   | 912.4323   | 328.473887(15) | 2.964*                          | 4.683e+06            | 10.61          | 5          | proposed spin-flip repump channel for the blue-slower sch... |
| 7s7p 3P <sub>2</sub> <sup>o</sup> – 7s8s 3S <sub>1</sub>            | 993.4946   | 993.2223   | 301.7555       | 6.025                           | 2.500e+07            | 8.793          | 45.9       | 993.5 nm                                                     |
| 7s6d 3D <sub>1</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup>            | 1428.6041  | 1428.2136  | 209.8499(6)    | 0.377                           | 3.300e+04            | 28.27          | 0.0231     | repumper                                                     |
| 7s7p 1P <sub>1</sub> <sup>o</sup> – 7s8s 3S <sub>1</sub>            | 1656.0662  | 1655.6140  | 181.0268       | 1.458*                          | 3.159e+05            | 8.793          | 0.572      | 1656 nm                                                      |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πc).

## <sup>223</sup>Ra hyperfine structure (*I* = 3/2)

| Level                             | A (MHz)   | B (MHz)    | Shift of each F level from the centre of gravity (MHz)         |
|-----------------------------------|-----------|------------|----------------------------------------------------------------|
| 7s7p 3P <sub>1</sub> <sup>o</sup> | 1202.1(6) | -470.2(12) | F=1/2: -3593.00 F=3/2: -731.90 F=5/2: +1685.60                 |
| 7s7p 3P <sub>2</sub> <sup>o</sup> | 700(3)    | 688(4)     | F=1/2: -2545.76 F=3/2: -2098.80 F=5/2: -780.11 F=7/2: +2270.93 |
| 7s7p 1P <sub>1</sub> <sup>o</sup> | -344.5(9) | 421.5(16)  | F=1/2: +1388.12 F=3/2: -77.00 F=5/2: -411.38                   |
| 7s7d 3D <sub>3</sub>              | 536(5)    | 74(3)      | F=3/2: -3169.26 F=5/2: -1885.68 F=7/2: -36.95 F=9/2: +2428.68  |

Measured A, B constants; sources in the data file.

## Isotope shifts

| Transition                                                          | λ vac (nm) | Isotopes | Shift (MHz) |
|---------------------------------------------------------------------|------------|----------|-------------|
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 208-214  | 8485(8)     |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 209-214  | 8313(8)     |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 210-214  | 6003(6)     |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 211-214  | 5522(4)     |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 212-214  | 3256(3)     |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 213-214  | 2167(3)     |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 220-214  | -21904(8)   |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 221-214  | -25945(8)   |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 222-214  | -28827(8)   |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 223-214  | -32453(9)   |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 224-214  | -35123(9)   |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 225-214  | -38888(10)  |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 226-214  | -41124(11)  |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 227-214  | -43937(12)  |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 228-214  | -46950(12)  |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 229-214  | -50066(15)  |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 230-214  | -53634(15)  |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 232-214  | -59585(17)  |
| 7s <sup>2</sup> 1S <sub>0</sub> – 7s7p 1P <sub>1</sub> <sup>o</sup> | 482.728    | 225-226  | 2236(15)    |
| 7s7p 3P <sub>2</sub> <sup>o</sup> – 7s7d 3D <sub>3</sub>            | 644.798    | 212-214  | -887(4)     |
| 7s7p 3P <sub>2</sub> <sup>o</sup> – 7s7d 3D <sub>3</sub>            | 644.798    | 221-214  | 7001(7)     |
| 7s7p 3P <sub>2</sub> <sup>o</sup> – 7s7d 3D <sub>3</sub>            | 644.798    | 222-214  | 7781(4)     |

Shift = ν(first isotope) – ν(second isotope).